

PROJECT TRICORDER

MK I · FIELD OPERATIONS MANUAL

On-device multimodal field instrument: inquiry, vision, image synthesis, neural speech in/out, offline translator (≥50 languages), audio scene analysis, avian species ID, flora/fauna classification, artificial horizon, NOAA and flight watch, append-only field log, and optional short-range tricorder-to-tricorder mesh.

Field	Value
Software version	0.5.4 (versionCode 32)
Application ID	com.projecttricorder.app
Platform	Android 8.0+ (API 26), arm64-v8a
Reference hardware	Samsung Galaxy Z Fold3 (Snapdragon 888, 12 GB)
Document date	4 September 2026
Classification	Unclassified — operator manual
Companion artifact	Tricorder-0.5.4-horizon-watch.apk
Supersedes	0.4.1 operations manual (2 September 2026)

NAVIGATION AND CERTIFICATION WARNING

This instrument is NOT certified by the FAA, NASA, EASA, ICAO, or any civil or military aviation or spaceflight authority. The artificial horizon, heading rose, accelerometer readout, barometer, and any derived attitude or heading display are uncertified consumer-sensor visualizations. They MUST NEVER be used as primary or backup navigation, attitude, or approach references for any aircraft, rotorcraft, UAV, or spacecraft. Do not use Project Tricorder to land, taxi, fly, dock, or station-keep any vehicle. Treat the horizon as a field curiosity and a demo of on-device IMU fusion only.

1. Purpose and scope

This manual is the single operator and maintainer reference for Project Tricorder Mk I through software 0.5.4. It covers installation of the Android application and the four local inference services, day-to-day operation of every control added since 0.4.1, theory of operations, mesh and watch services, design notes, a competitive matrix, and a fault tree.

It does not replace model licenses. Gemma, SmolVLM, Stable Diffusion, YAMNet, BirdNET, Kokoro, Coral/iNaturalist, and Google ML Kit language packs remain under their upstream terms. The operator must accept those terms before downloading weights.

Tricorder is an instrument first and a chatbot second. The design goal is a pocket logger that can look, listen, speak, translate, imagine, watch the sky and the weather, and write a timestamped book without a network account.

2. System overview

2.1 What the operator holds

Two cooperating halves:

- The APK — Kotlin WebView shell, phosphor CRT UI, on-device SpeechRecognizer, YAMNet and Task Vision in-process, ML Kit Translation, field log, Nearby mesh, NOAA/OpenSky watch, artificial horizon.
- Termux services — four localhost HTTP servers the APK treats as OpenAI-compatible backends: Gemma chat on 8080, SmolVLM vision on 8081, Stable Diffusion Turbo on 8188, Kokoro TTS on 5000.

The APK will open and show sensor chrome without Termux. Inquiry, live vision captions, image generation, and spoken replies in inquiry/sensor/sentry require the matching service. Audio scene analysis, BirdNET, flora/fauna, the translator, the horizon, and field logging do not require Termux.

2.2 Operating modes

Control	Mode	Primary path	Side effects
I Inquiry	CHATTING	Mic or type → Gemma :8080 → Kokoro :5000	IMAGE_GEN keywords route to SD :8188; “computer” easter egg
II Sensor	SCANNING	Camera JPEG → SmolVLM :8081 + flora TFLite	Auto-hold after caption; field log + snap
III Sentry	GUARDING	Motion / jerk → vision caption	Alert, field log + snap; mesh if Camp link on
IV Audio	LISTENING	Duty-cycle mic → YAMNet → BirdNET	Starts NOAA/flight watch if those boxes are on; mesh push
T Translator	XLAT	SpeechRecognizer → ML Kit → device TTS	AUTO swaps source/target after each spoken line
L Log review	BOOK	Slider over field.jsonl	Shows stored snaps and text
E Log entry	OFFICER	Mic/type → field log row	Spoken officer note
V Video log	REC	Front camera clip	Saved under files/logs/

2.3 What 0.5.x added

- 0.5.0 — Offline ML Kit translator, TRANSLATE lamp, pack fetch.
- 0.5.1 — SOURCE / TARGET language wheels and AUTO swap (default on, green).
- 0.5.2 — Translator re-arm after leaving Inquiry (destroys recognizer + TTS + ML Kit client).
- 0.5.3 — Keyboard-pop toggle, I-IV on one row, Demo Shell / wrapper chrome removed.
- 0.5.4 — Artificial horizon (idle + Audio), NOAA + OpenSky watch bound to Audio sense, mesh phrasing for remote hits.

3. Architecture

3.1 Process map

The APK process hosts WebView + JS tuple engine + Kotlin bridges. Termux is a separate Linux userspace. They meet only on 127.0.0.1. No cloud endpoint is required for the four generative backends. Translator packs and NOAA/OpenSky do use the public Internet when the operator opts in.

Layer	Implementation	Notes
Presentation	WebView + tricorder-ui-shell.html	CRT, matrix, sliders, lights, horizon
Intent / routing	JS TupleEngine	button seed + keyword hints → route_to
JS ↔ native	TricorderBridge	mic, Kokoro POST, audio, flora, log, Nearby, xlat, watch
Classifiers	TFLite Task Audio + Task Vision 0.4.4	YAMNet in APK; BirdNET and Coral fetched
Translator	ML Kit Translation + SpeechRecognizer + TTS	Offline after language packs download
Chat LLM	llama-server :8080	Gemma 3 4B Instruct GGUF
Vision VLM	llama-server :8081	SmolVLM2 500M + mmproj
Image gen	sd-server :8188	SD Turbo Q4; size/steps sliders
Speech out	Kokoro ONNX :5000	PocketPal-class graph
Watch	ContextWatch.kt	NWS + OpenSky while Audio is armed
Mesh	Nearby Connections P2P_STAR	JSON rows only, no media
Book	FieldLog JSONL	files/logs/field.jsonl

3.2 Local ports

Bind	Service	Required by
127.0.0.1:8080	llama-server chat (Gemma)	I Inquiry
127.0.0.1:8081	llama-server vision (SmoIVLM)	II Sensor, III Sentry
127.0.0.1:8188	sd-server	Imagine / IMAGE_GEN
127.0.0.1:5000	Kokoro HTTP	Spoken replies (not translator TTS)
0.0.0.0:8022	Termux sshd	Desk SSH only

3.3 Storage map

Path	Contents
app filesDir/logs/field.jsonl	Unified field book (audio, bird, flora, vision, sentry, xlat, context)
app filesDir/logs/audio.jsonl	YAMNet/BirdNET duty-cycle rows
app filesDir/models/	BirdNET and Coral TFLite packs
ML Kit model cache	Offline translate packs (~30 MB per language)
Termux ~/models or similar	GGUF, ONNX, SD checkpoint
/sdcard/Download/	Optional operator-side copies of packs

4. Theory of operations

4.1 Tuple routing

Every I Inquiry utterance is matched against canned tuples first (including the “computer” briefing). Unmatched text goes to Gemma. Keywords such as “show me / imagine / generate” route to sd-server with the current size and step sliders.

4.2 Inquiry path

Mic uses Android SpeechRecognizer. Keyboard pop is off by default (Settings → Keyboard pop). Spoken replies come from Kokoro :5000. The TRANSLATE path is a separate recognizer so Inquiry and Translator do not share a live session.

4.3 Sensor and Sentry

Sensor captures a frame, sends it to SmolVLM, and runs the Coral flora/fauna classifier on the same JPEG. After a caption the hold latch goes red so the loop does not flood Kokoro. Sentry uses motion/jerk; a hit writes description + snap + timestamp and can ride the mesh.

4.4 Audio path

Audio sense duty-cycles the microphone (listen/sleep sliders). YAMNet labels the scene. If a bird-like class fires and BirdNET weights are present, the same PCM is classified to species. Hits go to the field book. If Camp link is on, the JSON row is sent over Nearby.

4.5 Translator path

T arms a dedicated SpeechRecognizer in the source locale. Silence (onEndOfSpeech) ends the turn. ML Kit translates offline to the target language. Device TTS speaks the target line. If AUTO is green, source and target swap after TTS finishes so the other speaker can answer without touching the wheels. Press T again to disarm. Leaving T for Inquiry tears down recognizer, TTS, and the ML Kit client; returning to T creates a fresh session (0.5.2).

4.6 Watch path (Audio + Wi-Fi)

Arming IV also starts ContextWatch with whatever context_providers the operator enabled in Settings. NOAA is polled about every 180 s at the last GPS/network fix. OpenSky is polled about every 45 s in a ~25 km box. A new alert is logged, spoken, drawn on the CRT, and — if Camp link is up — pushed to the peer. Citizen and Waze checkboxes currently emit a one-time notice: those vendors have no keyless public feed.

4.7 Horizon path

While STANDBY or IV Audio is engaged, the CRT draws a sky/ground tape from accelerationIncludingGravity (pitch/roll) and a heading rose from device orientation. The numeric ACC/ROT line is hidden in those modes so the tape is readable. Sensor and Sentry keep the numeric line. See the certification warning on page 1.

4.8 Camp link path

Nearby Connections, strategy P2P_STAR, service id owned by the APK. Payload is JSON only (labels, species, context alerts). Media is not streamed. Both units must have Camp link on. First-time pairing may need Location and Nearby devices permissions.

5. Capabilities and limits

Capability	Where it runs	Network	Limit
Chat (Gemma 3 4B)	Termux :8080	None after weights	Seconds to tens of seconds per turn
Vision (SmolVLM2)	Termux :8081	None after weights	Hold recommended; continuous vision is screen-only
Imagine (SD Turbo)	Termux :8188	None after weights	Operator sliders 256–512, 4–8 steps; minutes on SD888
Kokoro speech	Termux :5000	None after ONNX	Full utterance then play; strip periods in long briefings
YAMNet	In APK	None	521 AudioSet classes
BirdNET	In APK after fetch	Fetch once	Needs YAMNet bird-like trigger
Flora/fauna	In APK after fetch	Fetch once	Coral iNaturalist; best on Sensor hold frame
Translator	ML Kit in APK	Pack download only	50+ languages; pair or all-packs fetch
NOAA watch	APK HTTPS	Wi-Fi/cell while Audio armed	Needs location fix + User-Agent
Flight watch	OpenSky HTTPS	Wi-Fi/cell while Audio armed	Anonymous rate limits; ~25 km box
Camp mesh	Nearby	No Internet	JSON only; tens of meters typical
Horizon	IMU in APK	None	NOT a flight instrument

6. Installation

6.1 Sideload the APK

Uninstall any older com.projecttricorder.app build first (debug signature). Install Tricorder-0.5.4-horizon-watch.apk. Grant Camera, Microphone, Location, and Nearby devices when asked.

6.2 Termux stack

Use the existing ~/bin/tricorder start helper so sshd :8022, chat :8080, vision :8081, image :8188, and tts :5000 come up together. Desk SSH dropping should not kill the stack if started on-device.

6.3 In-APK fetches

- BirdNET — Settings fetch button; weights land under the app filesDir.
- Flora/fauna — same pattern; Coral test_data MobileNet packs.
- Translator — “Fetch source+target pair only” (~60 MB) or “Fetch ALL translator packs.” Stay on Wi-Fi. Status prints on the CRT.

6.4 Permissions that matter for watch and mesh

- Location — last-known fix for NWS point alerts and the OpenSky box.
- Nearby devices + Location — Camp link.
- Microphone — Audio, Inquiry mic, Translator.
- Internet — pack fetch, NOAA, OpenSky only.

7. Guide to operations

7.1 Lights

IDLE / ACTIVE / ALERT are mode lamps. CHAT, VISION, IMAGE, VOICE probe the four Termux ports (green idle, amber busy, red down). TRANSLATE is independent: green listening, red thinking, amber speaking the target line.

7.2 Inquiry

Tap I. Speak with Mic. Keyboard stays down unless Keyboard pop is on (under Imagine sliders). Text clears after send. “Computer” triggers the Daystrom briefing (periods stripped before Kokoro).

7.3 Sensor / Sentry / Hold

II captures and captions; flora classifier runs on the same frame. Hold is a toggle — red holds the last frame and will not re-describe until released. Continuous machine vision writes the CRT only and does not call Kokoro.

7.4 Audio sense

IV arms YAMNet. Adjust listen/sleep sliders. Enable BirdNET fetch once. Check NOAA and/or flight_overhead in Settings if those watches are wanted. They only run while IV is engaged and a network path exists.

7.5 Universal translator

Tap T. SOURCE and TARGET open a scroll wheel of ML Kit languages. AUTO defaults green: after each spoken translation the two languages swap. Red AUTO is one-way. Fetch at least the working pair before the first field use. Press T again to stop.

7.6 Imagine sliders

Size 256 / 384 / 512 and steps 4 / 6 / 8. On this SoC, 512 × 4 is the known-good speed/quality bargain; 256 at low steps tiles. Change sliders between images; they apply to the next request.

7.7 Keyboard pop

Default off. Operators who live on the mic should leave it off. On restores focus-and-raise-keyboard for Inquiry and officer log.

7.8 Log review, officer entry, video log

L + slider walks field.jsonl. E dictates an officer line into the book. V records a front-camera clip. All timestamps are device clock; GPS is attached when a fix exists.

7.9 Horizon

Visible in STANDBY and IV. Heading is degrees from device orientation; P and R are pitch and roll in degrees from gravity. Recalibrate by holding the unit level if the tape looks biased. Do not fly on it.

8. NOAA, flight watch, and tricorder-to-tricorder comms

8.1 Design intent

Audio sense is the duty officer. The microphone already wakes on a schedule; the same armed state is the right time to ask the outside world whether the weather office or the airspace has something the operator should hear. A pair of units — one in the tent, one at the fishing hole — share those hits the same way they share a growled bear: JSON over Nearby, spoken on the far CRT.

8.2 Enabling the watches

- Open Settings → Context providers.
- Set `weather_alerts_noaa.enabled` and/or `flight_overhead.enabled`.
- Optional: set the NWS User-Agent to an address you control (NWS requires a contact string).
- Turn Wi-Fi or cell on. Confirm a location fix (go outside if the last fix is stale).
- Turn Camp link on at both units if the far unit should hear the hit.
- Tap IV Audio on the unit that will sit watch.

8.3 What a local hit sounds like

Local tricorder detected weather event at HH:MM. Details follow. <NWS headline — area>.

Local tricorder detected flight event at HH:MM. Details follow. <callsign altitude speed; ...>.

8.4 What the remote unit speaks

Remote tricorder unit detected weather event at HH:MM. Details follow. <same details>.

The same template is used for flight and for audio/bird rows that arrive over the mesh. The far operator should hear the source class, the clock, then the payload — not a raw JSON dump.

8.5 Camp link procedure

- Both phones: Camp link on (Settings / Audio aux).
- Keep them in radio range (same campsite, unobstructed, typically tens of meters).
- CRT should note LINK advertising / connected. If it never links: toggle link off/on, confirm Location + Nearby permissions, disable battery restrictions for the APK.
- Only JSON rows move. Pictures from Sensor/Sentry stay on the originating unit's field book unless copied by hand.

8.6 Failure modes

- Checkbox on, Audio off — no poll. Arm IV.
- Audio on, Wi-Fi off — YAMNet still runs; NOAA/OpenSky stay quiet.
- No GPS — NWS point query and the OpenSky box have nothing to center on.
- OpenSky HTTP 429 — anonymous quota; the poller swallows the error and tries later.
- NWS 403 — User-Agent rejected; put a real contact in Settings.
- Citizen / Waze checked — one notice, no feed. Do not expect crime or traffic alerts.

9. Design notes

Translator uses ML Kit rather than a Termux Python stack so speech-to-speech stays inside the APK and survives a Termux crash. AUTO swap lives in Kotlin after TTS onDone so the next listen already uses the new source locale.

Watch services are deliberately not a background Android foreground-service yet. They ride Audio sense so the operator has an obvious armed/disarmed control and so a unit in a pocket is not quietly hitting NWS all day.

The horizon is a 2-D tape, not a 3-D GL scene, so it stays cheap on the WebView thread and readable at 72 px.

10. Competitive matrix (indicative)

Y = ships as a first-class control. P = partial or plugin. N = not in the product.

Capability	Tricorder 0.5.4	PocketPal	ChatGPT app	Google Translate	Seek / iNaturalist	Merlin
Offline LLM chat	Y	Y	N	N	N	N
Offline VLM	Y	P	N	N	N	N
On-device SD image	Y	N	N	N	N	N
Neural TTS (Kokoro)	Y	Y	P	P	N	N
Offline voice translate	Y	N	N	Y	N	N
YAMNet scene + log	Y	N	N	N	N	N
BirdNET species	Y	N	N	N	N	Y
Flora/fauna TFLite	Y	N	N	N	Y	N
Field book + snaps	Y	N	N	N	P	P
Camp mesh alerts	Y	N	N	N	N	N
NOAA + OpenSky watch	Y	N	N	N	N	N
Artificial horizon	Y*	N	N	N	N	N
No account required	Y	Y	N	P	P	P

* Horizon is uncertified. See the warning on page 1.

11. Troubleshooting

Symptom	Likely cause	Action
Inquiry Awaiting query	Send path / stale focus	0.4.1+; confirm mic result fires Send; keyboard pop off is fine
Translator beeps on return	Recognizer left busy	Use 0.5.2+; wait one second after leaving I before T
Translator silent	Packs missing	Fetch pair; lamp should go green then amber
Audio armed, no labels	fntScore / hold latch / duty	Confirm 0.3.2+ audio path; Hold green; raise dwell
Flora never prints	No pack or no hold frame	Fetch Coral nets; use Sensor hold so the JPEG is classified
Vision lamp red	8081 down	~/bin/tricorder status; restart vision
Image 400 / tiles	Size 256 or 2 steps	512 × 4 or 384 × 6
Kokoro mute	:5000 down or NaN WAV	Check tts.log; strip "U.S.S." style periods in long text
NOAA never fires	Audio off, no fix, no net	IV + Wi-Fi + location + checkbox
No remote speech	Camp link off or range	Link on both units; stay in the same clearing
Horizon frozen at 0	Motion permission / sensors idle	Tilt the unit; accept motion if the OS prompts
Services die after desk SSH	Started in that SSH session	Run ~/bin/tricorder start on the phone

12. Maintenance

Keep one sideloaded APK. Do not mix 0.4.x and 0.5.x data assumptions without exporting field.jsonl first. After a Play Services update, re-fetch a translator pair if XLAT throws on first use. NWS and OpenSky are courtesy feeds — expect outages. Review field.jsonl; it grows without bound.

Thermal: SD Turbo and Gemma on a Fold3 will heat the inner display. Pause Imagine in a pocket. Watch duty (Audio + radios + Nearby) is lighter than Imagine but still keep the unit ventilated if it sits in a tent for hours.

13. Document control

Rev	Date	Notes
0.4.1	2 Sep 2026	First bound operations manual (inquiry fix, field log, camp link).
0.5.4	4 Sep 2026	Translator (50+ languages, AUTO swap), horizon, NOAA/OpenSky watch, mesh phrasing, keyboard toggle, I–IV single row. FAA/NASA navigation disclaimer added.

End of manual.